METRIC

MIL-C-50872C (AR)
AMENDMENT 3
8 September 2000
SUPERSEDING
AMENDMENT 2
24 September 1993

MILITARY SPECIFICATION

CARTRIDGE, 40MM, HEDP, M433
LOADING, ASSEMBLING AND PACKAGING

MIL-C-50872C was inactivated on 8 February 1996 for new design.

This amendment forms a part of Military Specification MIL-C-50872C (AR), dated 22 July 1981, and is approved for use by the U.S. Army Armament, Research, Development, and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

* 3.4, Delete in its entirety.

PAGE 5

* 4.4a, Delete in its entirety and substitute:

"a. M118 Cartridge case loading assemblies metal parts from one lot interfix number from one manufacturer."

* 4.4f, Delete in its entirety and substitute:

"f. RDX or Composition A-5 from one lot interfix number from one manufacturer."

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* 4.4.2.1:

Under Critical defects, delete "None defined" and substitute: "1. X-ray examination 100% 3.9 4.4.3.4"

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Add Major 104 as follows:

"104. X-ray examination 100% 3.9 4.4.3.4"

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* 4.4.2.5:

Delete critical 1 in its entirety.

Add new Major 103 as follows:

"103. Chamber gage failure (after complete cure) 100% 3.2"

4.4.2.5, Delete Major 102 in its entirety and substitute:

"102. Air pressure 4.5.7 4.5.7 3.2 4.5.7"

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* 4.4.3.4, Delete in its entirety and substitute the following:

"4.4.3.4 X-ray examination of fuze assemblies prior to assembling to cup and liner assembly. Each fuze assembly prior to assembling to cup and liner assembly shall be x-rayed. The inspection shall be conducted in two (2) planes; from the top of the fuze assembly down; and from the side. Each view shall be read by two (2) independent inspectors. If any fuze shows evidence of being improperly assembled or having missing parts, it shall be rejected. Observation shall be made for the following critical defects:

a. Setback pin not seated in bottom of counterbore of escapement housing.
b. Setback pin missing.
c. Setback pin spring missing.
d. Detent missing.
e. Detent spring missing.
f. Detent end not in contact or seated within root of teeth of rotor gear.

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g. Pinion assembly missing.
h. Verge assembly missing.
i. Rotor gear assembly in armed or partially armed position.
j. A major defect is defined as any defect not specified as critical above.

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* 4.4.3.7, Delete ". . .the Government inspector. . ." and substitute ". . . the Government or contractor. . ."

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* 4.4.3.8, Delete ". . .the Government inspector. . ." and substitute ". . . the Government or contractor. . ."

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Add new paragraph 4.5.4.1:

"4.5.4.1 Alternate method for specific gravity determination.

Prior to assembly, weigh empty body assembly with liner assembly and retainer in air (grams). Record the weight as "A".

Prior to assembly, weigh the same empty body assembly with liner assembly and retainer submerged in water (grams). Record the weight as "B".

Dry the components and load with explosive composition using normal production methods.

Weigh the loaded assembly in air(grams). Record the weight as "C".

Weigh the loaded assembly submerged in water (grams). Record the weight as "D".

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Calculate the specific gravity of charge as follows:

$$\text{Sp. Gr.} = \frac{C-A}{(C-A) - (D-B)}$$

4.5.7, Add:

"Fourteen (14) samples shall be selected from every thirty (30) minutes of production. The last sample shall be air tested. If it is acceptable, the thirty (30) minutes of production shall be considered acceptable; and the samples shall be returned to the lot. If the sample is unacceptable, the remaining thirteen (13) samples shall be air tested. If no failure occurs, the thirty (30) minutes of production shall be considered acceptable. If one (1) or more failures occur, the thirty (30) minutes of production shall be rejected and segregated from the lot."

4.5.9, delete ". . .70°F + 5°. . ." and substitute ".. .70°± 5°. . ."

* 4.5.9, Line 11, Delete "The velocity shall be measured at twenty (20) feet from the muzzle of the launcher, using two sets of screens each set of screens having an electronic counter."

And substitute:

"The velocity shall be measured at twenty (20) feet from the muzzle of the launcher. If an approved radar system is not used, then the monitoring devices and timer recorders shall be a dual setup, using two sets of screens, each set of screens having an electronic counter."

The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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